A 24 held-out phantoms (T_1s uniform over the T_1-plate range, as in training)

Method	Memory	Budget	Mean scan time	MAPE % [95% CI]	Scans < 5%	Worst scan MAPE %
RL	**TI histogram**	**560 s**	**541 s**	**2.93 [2.58, 3.30]**	**95.8%**	**5.13**
RL	none	560 s	548 s	4.16 [3.41, 4.99]	66.7%	9.50
RL	LSTM†	560 s	525 s	7.31 [6.10, 8.65]	25.0%	15.02
Fixed, distilled from RL policy§	–	560 s	549 s	3.34 [2.30, 4.86]	91.7%	17.87
Fixed log TI grid	–	560 s	504 s	6.04 [5.36, 6.70]	25.0%	10.29
Cramér-Rao optimal	–	560 s	558 s	7.74 [7.08, 8.42]	8.3%	11.72
RL	none	240 s	227 s	4.62 [4.06, 5.19]	62.5%	7.74

B 96 clustered-T_1 phantoms (T_1s within ±15% of a random centre; policy not retrained)

Method	Memory	Budget	Mean scan time	MAPE % [95% CI]	Scans < 5%	Worst scan MAPE %
RL	**TI histogram**	**560 s**	**545 s**	**2.76 [2.58, 2.95]**	**96.9%**	**5.35**
Fixed, distilled from RL policy§	–	560 s	549 s	12.20 [8.02, 16.94]	78.1%	91.47
Fixed log TI grid	–	560 s	504 s	8.31 [7.55, 9.10]	18.8%	22.68
Cramér-Rao optimal	–	560 s	558 s	8.53 [7.95, 9.16]	10.4%	18.56

† Undertrained: the LSTM ran 4.6× slower, so within the same 9 h budget it took 7.3× fewer full-Bloch steps. Its best checkpoint predates full-Bloch training, so this is not a fair test of recurrent memory.

§ One fixed schedule: per-block median TI and TR of the TI-histogram policy's schedules on 24 validation scans (13 blocks, 549 s).